Week 13 Homework

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In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred () proof problems are eligible for inclusion in the proof portfolio assignment. ## Exercises

Exercise

Find out what the Chinese Postman Problem is. Describe it, and give examples. What does the Chinese Postman Problem have to do with Ribonucleic acid (RNA)?

Exercise

Show that if you delete any vertex and it's incident edges from the Petersen graph then the resulting graph is Hamiltonian.

Exercise

From the text, Exercise 4.3.

Exercise

From the text, Exercise 4.4.

Exercise

A **Hamiltonian path** is a path that begins and ends at different vertices but visits each vertex of a graph exactly once. If a graph has a Hamiltonian cycle, must it have a Hamiltonian path? Explain why or why not. If a graph has a Hamiltonian path must it have a Hamiltonian cycle? Explain why or why not.

Exercise

What are the Worpitzsky numbers? What do they count? What is the recursive formula for these numbers? Give an example of something they can be used to count.

Exercise

From the text, Exercise 5.4.

Exercise

Find the coefficient of x, x^2, x^3, and x^4 in the Taylor expansion of

$$\frac{1}{(1-x)}\frac{1}{(1-x^2)}\frac{1}{(1-x^3)}\cdots$$

and compare to $p(1)$, $p(2)$, $p(3)$, $p(4)$. Do you want to make a conjecture? (Hint: find the taylor expansion, by multiplying together each of the fractions Taylor expansion).

Exercise

Find the definition of Stirling numbers of the first kind. What recursion relationship do these numbers satisfy? Give examples of the permutations counted by $s(5,2)$ and $s(6,3)$, where $s(n,k)$ is a Stirling number of the first kind.

Proofs

Proof

Let $s(n,k)$ denote the signless Stirling numbers of the first kind. Using the recursive formulat $s(n,k) = s(n-1,k-1) + (n-1)s(n-1,k)$, prove that

$$s(n,n-1) = \binom{n}{2}$$

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Proof

Let $s(n,k)$ denote the signless Stirling numbers of the first kind. Using the recursive formulat $s(n,k) = s(n-1,k-1) + (n-1)s(n-1,k)$, prove that

$$s(n,2) = (n-1)!\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1}\right).$$

Proof

Prove

$$\frac{\binom{2n}{2}\binom{2n-2}{2}\binom{2n-4}{2}\cdots\binom{2}{2}}{n!} = \frac{(2n)!}{2^n \cdot n!}$$

Proof

Prove

$$\frac{\binom{m\cdot n}{n}\binom{(m-1)n}{n}\binom{(m-2)n}{n}\cdots\binom{n}{n}}{m!} = \frac{(m\cdot n)!}{(n!)^m \cdot m!}$$

Proof

Let $s(n,k)$ denote the signless Stirling numbers of the first kind. Use the recursive formula $s(n,k) = s(n-1,k-1) + (n-1)s(n-1,k)$ to find a formula for $s(n,n-2)$ and prove it.

Proof

Suppose G is a graph with n vertices, each of which has degree $d \geq (n-1)/2$. Prove that G contains a Hamiltonian path. Hint: Add an extra vertex to G and connect it to all the other vertices. What does Dirac's Theorem say about this new graph.)

Proof

Prove $\sum_{r=0}^{n}\binom{n}{r}2^{2n-2r} = 5^n$